

Replication Report: (title not recorded)

OSTI 2396968 | Coverage 9/10, Agreement 6/10

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1 Source paper

- **Title:** (title not recorded)
- **Authors:** Not recorded
- **Year:** n/a
- **Domain:** Not recorded
- **Identifier:** OSTI 2396968

2 Replication objective

The central claim of the paper concerns not recorded. Our objective was to reproduce the head-line numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72h, commodity GPU/CPU compute, no author contact). A replication plan is archived at `2396968-Latent-Stochastic-Differential-Equations-for-Modeling/replication_plan.pdf`.

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper’s computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

This paper went through the Tier-Lift pipeline; supplementary notes at `.openclaw/workspace/24h-progress/tie`

Paper-directory README excerpt

Latent Stochastic Differential Equations for Modeling Quasar Variability and Inferring Black Hole Properties

- **OSTI ID:** 2396968
- **Rank:** #27
- **Replication Score:** 9/10 (paper); **v1** (1-band, deprecated) 4-6/10; **v2** (paper-faithful, 6-band, A100) 7/10
- **v2 report:** replication/v2_faithful/report/report.pdf

- ****v2 training:**** 18.08 h on uicgpu A100, 8 epochs, 20k curves, 917,594 params, full 6-band + Sim5 GR + 9-param Cholesky head
- ****v2 headline:**** LC RMSE 0.1091 mag (paper 0.0959, 1.14 $\times$); 1 / 2 / 3 coverage 73.3/94.5/99.0% (paper ~70/95/99.7%); per-parameter ranking matches paper exactly
- ****Open-Source Tools:**** Yes (PyTorch + torchsde)
- ****Code Repository:**** No (paper); this directory contains a from-scratch simplified replication

What was replicated

A single-band simplification of Fagin et al. (2024): encoder $\rightarrow$ latent Itô SDE (learned prior drift, diagonal diffusion) $\rightarrow$ decoder, trained with reconstruction NLL plus Girsanov KL via ‘torchsde.sdeint(..., logqp=True)’¹. We skip the 6-band LSST pipeline, the physical accretion-disk transfer functions, and the 9-parameter black-hole inference head.

Quantitative summary

Model	RMSE (mag)	MAE (mag)	NLL		Paper latent-SDE (6-band)	0.0959	0.0695
-1.14		Paper GPR (Matern- $\frac{1}{2}$, 6-band)	0.0978	0.0711	-1.01		**Ours latent-SDE (1-band, 512 train)**
0.198	0.149	+0.49		**Ours GPR (Matern-$\frac{1}{2}$, LOO, 1-band)**	0.048	0.034	-1.85

The gap to the paper is driven by (i) single-band setup — the paper’s edge over GPR largely comes from sharing information across six bands, (ii) a $\sim 2000\times$ smaller training set, (iii) an $\sim 17\times$ smaller model, and (iv) $\sim 3000\times$ less training compute (160 s CPU vs ~ 6 weeks V100).

Layout

[code block omitted]

How to reproduce (CPU, ~ 3 min total)

[code block omitted]

Status

- [x] Paper reviewed
- [x] Code/tools identified (PyTorch + torchsde)
- [x] Code implemented (from scratch — no public repo)
- [x] Simplified results reproduced (single-band DRW)
- [] Full-scale results validated against paper (out of scope: 6 weeks V100)

Notes / Caveats

- ‘torchsde.sdeint(..., logqp=True)’ is exactly the Li-et-al. 2020 / Fagin-et-al. formulation for the Girsanov KL between posterior and prior SDE paths.

[¹textit{README truncated}]

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper’s resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- Not recorded.

5 What was missing or incomplete

- Not recorded.

The dominant blocker categories for this paper, mapped onto the meta-paper’s 9-category friction taxonomy (§4), are listed in §?? below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. Not recorded.

7 Score and friction-point tagging

- **Coverage:** 9/10 — Not recorded.
- **Agreement:** 6/10 — Not recorded.

Friction points encountered (taxonomy tags):

- None recorded.

8 Follow-on questions

- Not recorded.

9 Reproducibility pointers

- **Paper directory:** 2396968-Latent-Stochastic-Differential-Equations-for-Modeling
- **Replication artefacts:**
 - /Users/stevens/Dropbox/REPLICATE-PROJECT/2396968-Latent-Stochastic-Differential-Equations-for-Modeling
 - /Users/stevens/Dropbox/REPLICATE-PROJECT/2396968-Latent-Stochastic-Differential-Equations-for-Modeling
 - /Users/stevens/Dropbox/REPLICATE-PROJECT/2396968-Latent-Stochastic-Differential-Equations-for-Modeling
- **Replication-dir hint from JSONL:** /Users/stevens/Dropbox/REPLICATE-PROJECT/2396968-Latent-Stochastic-Differential-Equations-for-Modeling
- **Status:** tier_lift_v2_2026_04_27

To re-run, change directory into the paper directory above and consult its `README.md` for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.